MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ADVANCED SCHOTTKY TTL, QUAD 2-PORT REGISTERS, CASCADABLE, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Advanced Schottky TTL, Quad 2-Port register microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Quad 2-Port cascadable register with both inverted and non-inverted outputs
02	Quad 2-Port cascadable register with non-inverted outputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (See MIL-M-38510, appendix C)
R	D-8 (20-lead, 1/4" x 1 1/16"), dual-in-line package
S	F-9 (20-lead, 1/4" x 1/2"), flat package
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.2 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	
Device type 01	209 mW
Device type 02	187 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	2/
Case R	(See MIL-M-38510, appendix C)
Case S	

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).
2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Case E	(See MIL-M-38510, appendix C)
Case F	
Case 2	0.08°C/W
Junction temperature (T_J)	+175°C 3/

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.8 V dc
Normalized fanout (each output) 4/	
Low logic level	33 maximum
High logic level	50 maximum
Case operating temperature range (T_C)	-55°C to +125°C
Width of clock pulse high:	
Device type 01, 02	4.0 ns minimum
Width of clock pulse low:	
Device type 01, 02	7.0 ns minimum
Setup time in high to clock pulse:	
Device type 01, 02	4.5 ns minimum
Setup time in low to clock pulse:	
Device type 01, 02	4.5 ns minimum
Setup time S high to clock pulse:	
Device type 01	10.5 minimum
Device type 02	9.5 minimum
Setup time S low to clock pulse:	
Device type 01	10.5 minimum
Device type 02	9.5 minimum
Hold time in high to clock pulse:	
Device type 01, 02	1.5 ns minimum
Hold time in low to clock pulse:	
Device type 01, 02	1.5 ns minimum
Hold time S high to clock pulse:	
Device type 01, 02	0 ns minimum
Hold time S low to clock pulse:	
Device type 01, 02	0 ns minimum

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.
4/ The device shall fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 1.

3.2.3 Logic diagram. The logic diagram shall be as specified on figure 2.

3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 12 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

   (1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

   (2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits		Unit
				Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -1.0 mA V_{IL} = 0.8 V, V_{IH} = 2.0 V	All	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 20 mA V_{IL} = 0.8 V, V_{IH} = 2.0 V	All		.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA T_C = +25°C	All		-1.2	V
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	All		20	µA
High-level input current	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 7.0 V	All		100	µA
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.5 V	All	-.03	-.60	mA
Short circuit output current 1/	I_{OS}	V_{CC} = 5.5 V, V_{IN} = 0 V	All	-60	-150	mA
Output drive	I_{OD}	V_{CC} = 4.5 V, V_{OUT} = 2.5 V, V_{IN} = 5.5 V	All	60		mA
Supply current	I_{CC}	V_{CC} = 5.5 V, V_{IN} = GND	01		38	mA
			02		34	
Propagation delay time, low to high level		V_{CC} = 5.0 V, C_L = 50 pF ±10% R_L = 500Ω				
Cp to Q, $\overline{Q}$	t_{PLH}		01	1.5	9.5	ns
Cp to Q	t_{PLH}		02	1.5	9.5	ns
Propagation delay time, high to low level		V_{CC} = 5.0 V, C_L = 50 pF ±10% R_L = 500Ω				
Cp to Q, $\overline{Q}$	t_{PHL}		01	1.5	11.5	ns
Cp to Q	t_{PHL}		02	1.5	11.5	ns
Input clock frequency	f_{MAX}		All	80		MHz

1/ Not more than one output should be shorted at a time.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1,2,3	1,2,3
Final electrical test parameters (method 5004)	1*,2,3,9	1*,2,3,7, 9
Group A test requirements (method 5005)	1,2,3,7, 8,9,10,11	1,2,3,7, 8,9
Group B test requirements (method 5005) subgroup 5	1,2,3,7, 8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

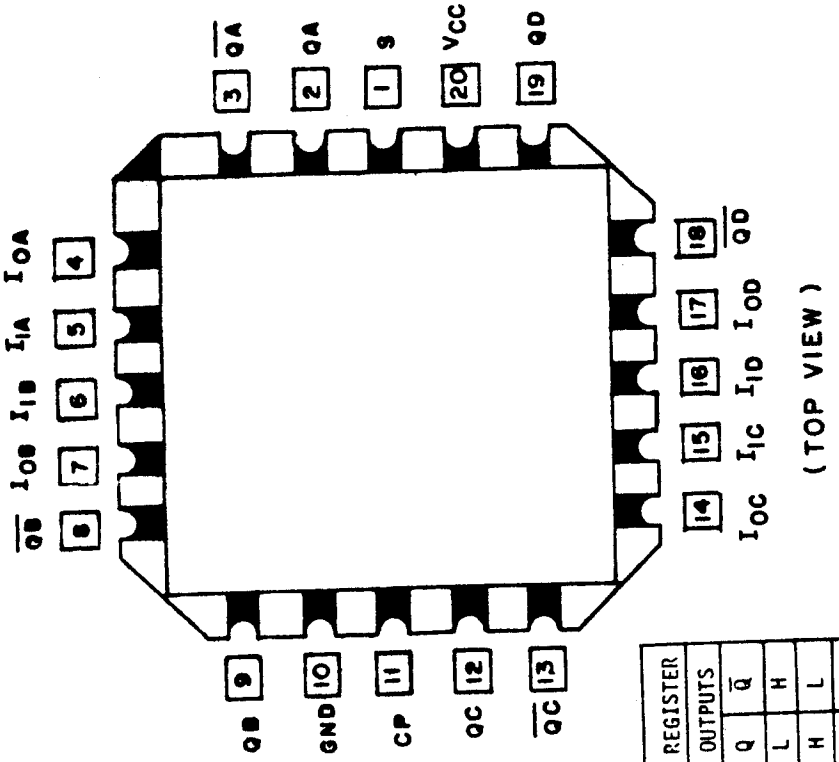
c. Subgroups 3 and 4 shall be added to group C inspection for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

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Device type 01

Case 2

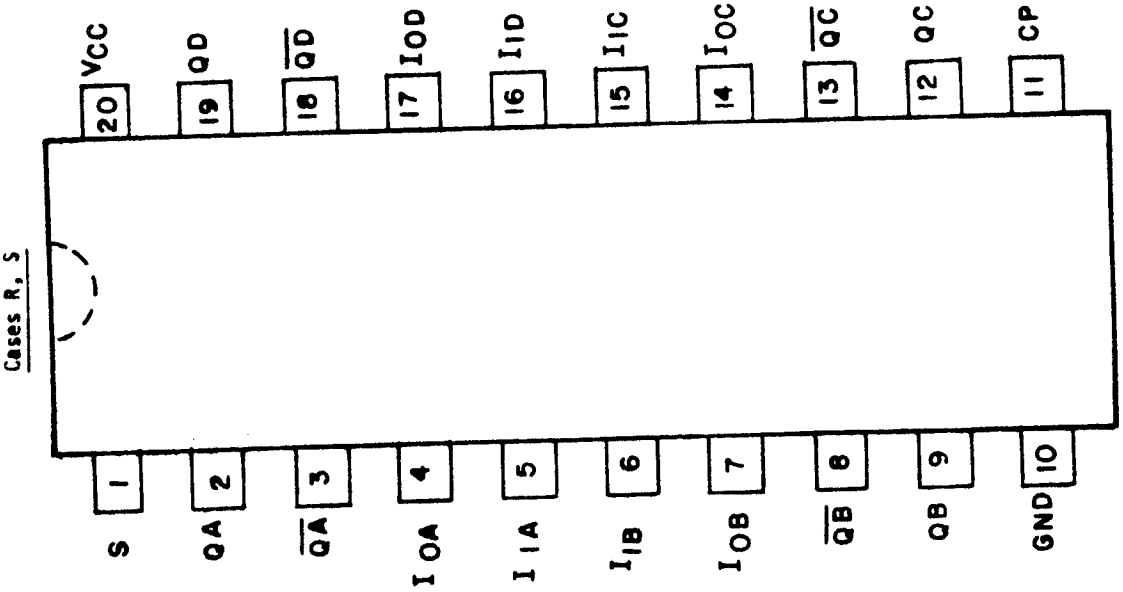


TRUTH TABLE EACH REGISTER					
INPUTS				OUTPUTS	
S	I_0	I_1		Q	$\overline{Q}$
I	I	X		L	H
I	h	X		H	L
h	X	I		L	H
h	X	h		H	L

I = LOW voltage level one setup time prior to the LOW to HIGH clock transition
h = HIGH voltage level one setup time prior to the LOW to HIGH clock transition
L = LOW voltage level
H = HIGH voltage level
X = Immaterial

FIGURE 1. Terminal connections and truth tables.

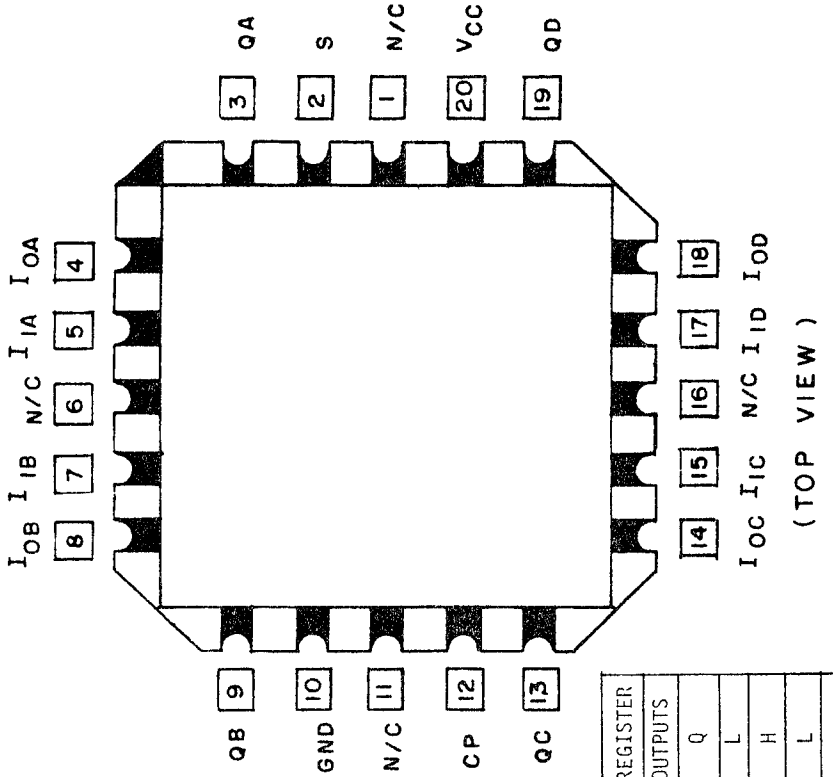
Cases R, S



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Device type 02

Case 2



TRUTH TABLE EACH REGISTER				
INPUTS			OUTPUTS	
S	I_0	I_1	Q	
I	I	X	L	
I	h	X	H	
h	X	I	L	
h	X	h	H	

I = LOW voltage level one setup time prior to the LOW to HIGH clock transition
h = HIGH voltage level one setup time prior to the LOW to HIGH clock transition
L = LOW voltage level
H = HIGH voltage level
X = Immaterial

Cases E, F

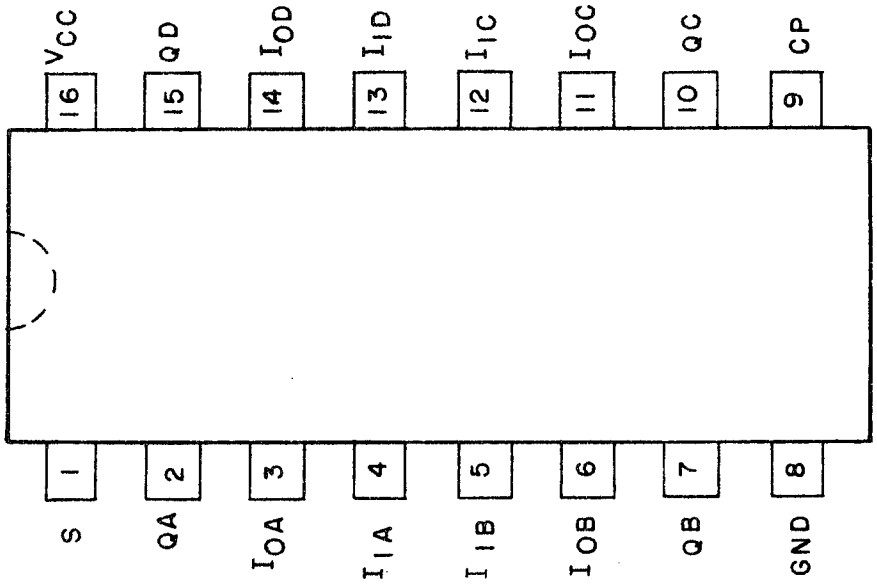
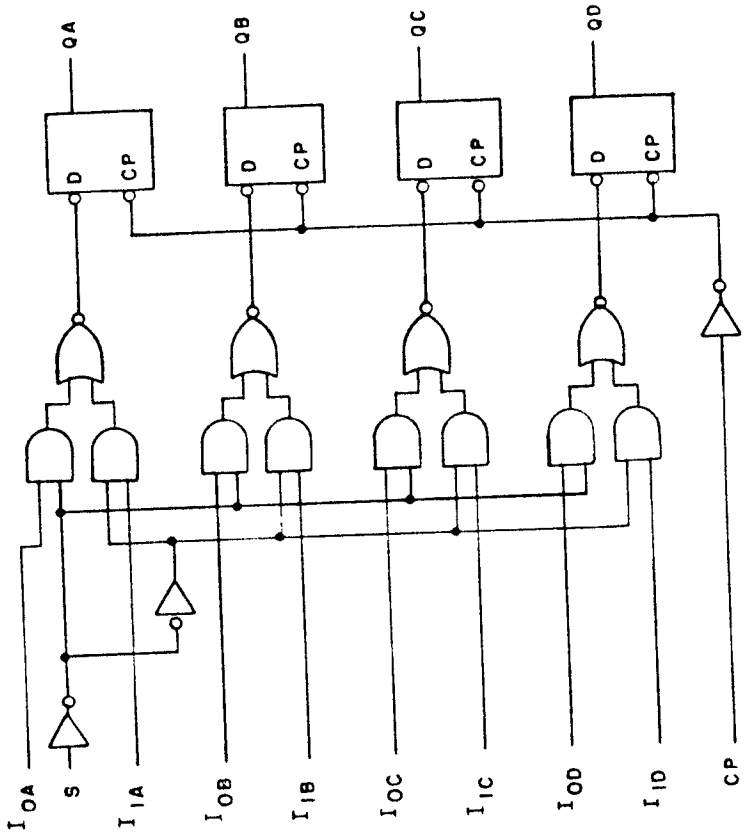


FIGURE 1. Terminal connections and truth tables - Continued.

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Device type 02



Device type 01

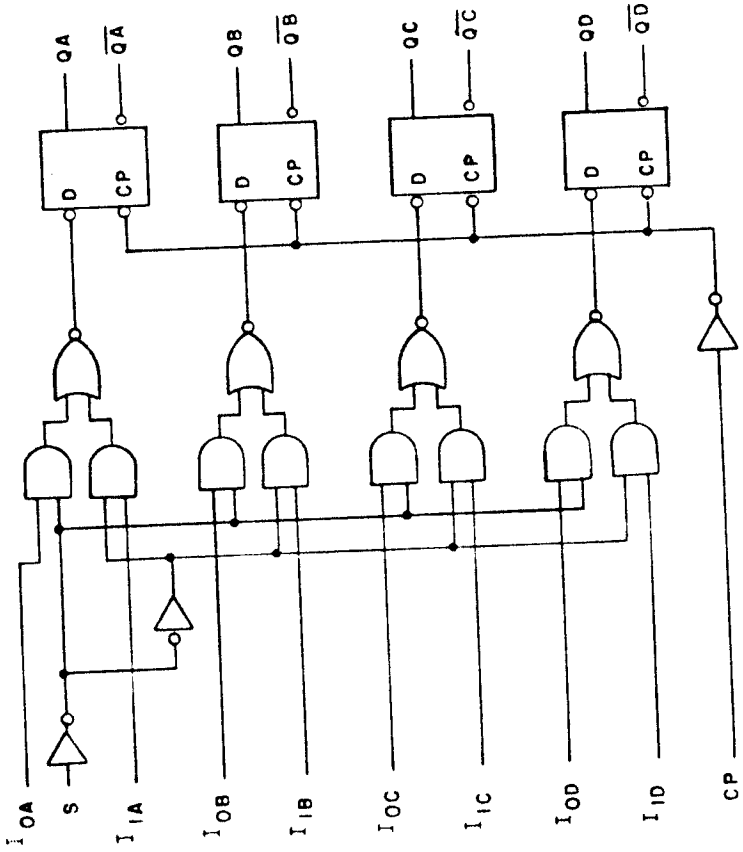


FIGURE 2. Logic diagrams.

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Device type 01

Cases R, S, 2

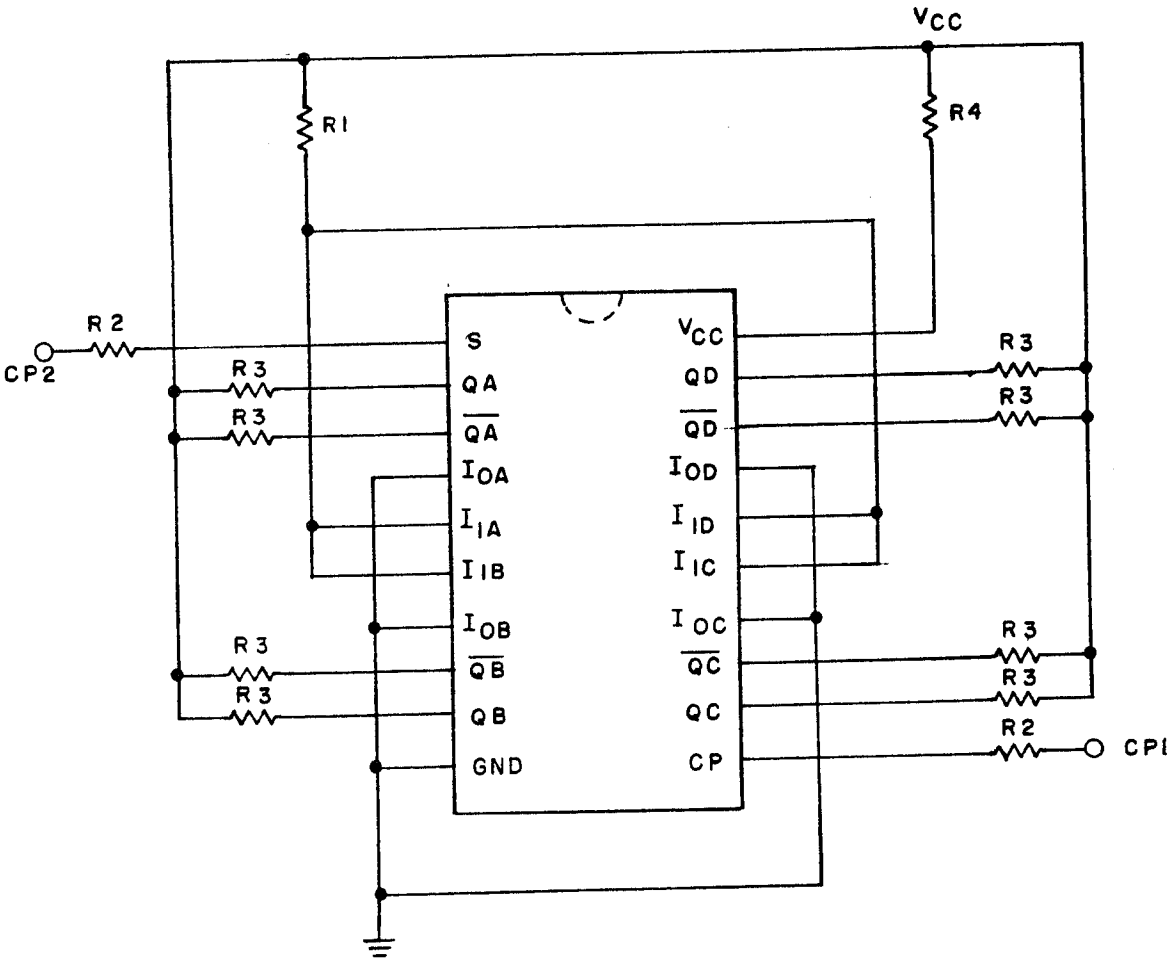
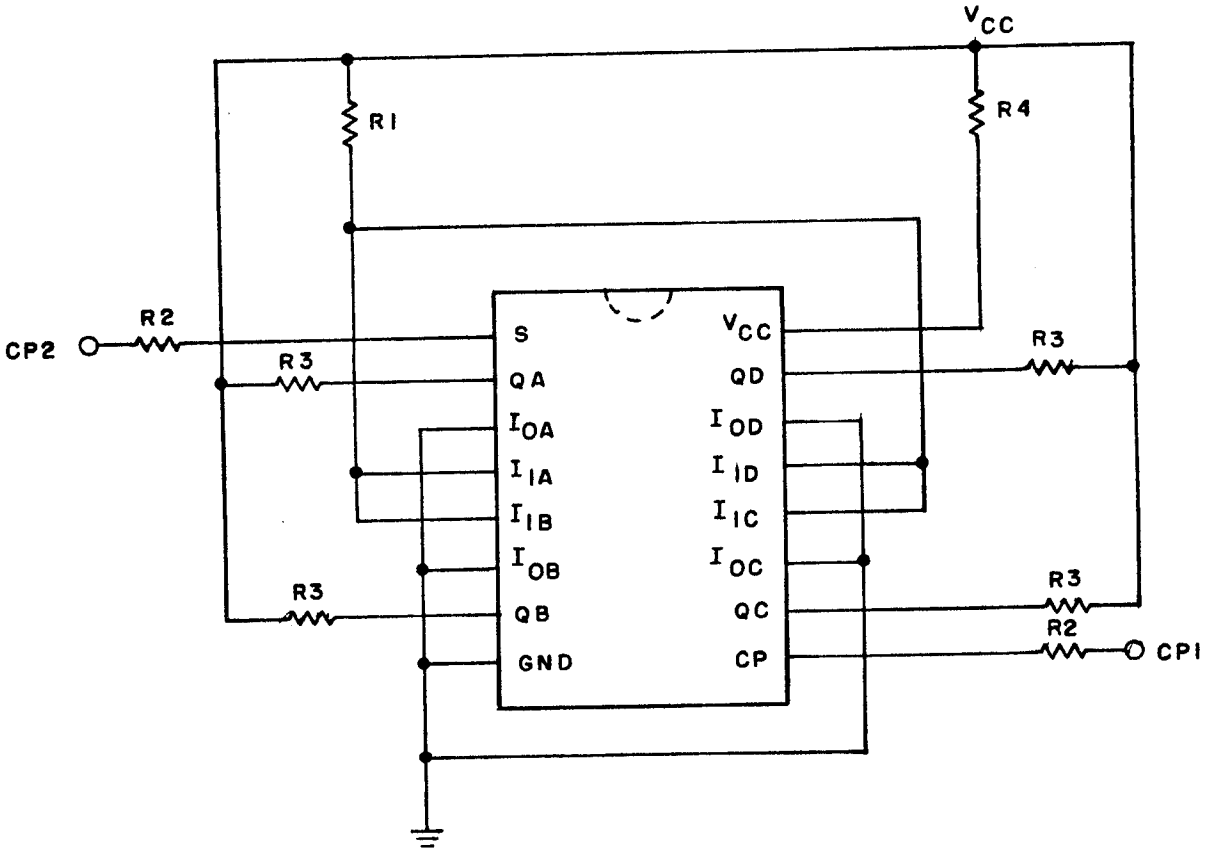


FIGURE 3. Burn-in and life-test circuit.

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Device type 02

Cases E, F, 2



NOTES:
1. CP 1 = 100 kHz ±50% square wave; duty cycle = 50% ±15%; V_{IL} = -0.5 V minimum to +0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. CP 2 = 50 kHz ±50% square wave synchronized with CP 1. All other conditions are the same as CP 1.
3. R1 = 1K ±5%, R2 = 50 ±5%, R3 = 270 ±5%.
4. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device terminal.

FIGURE 3. Burn-in and life-test circuit - Continued.

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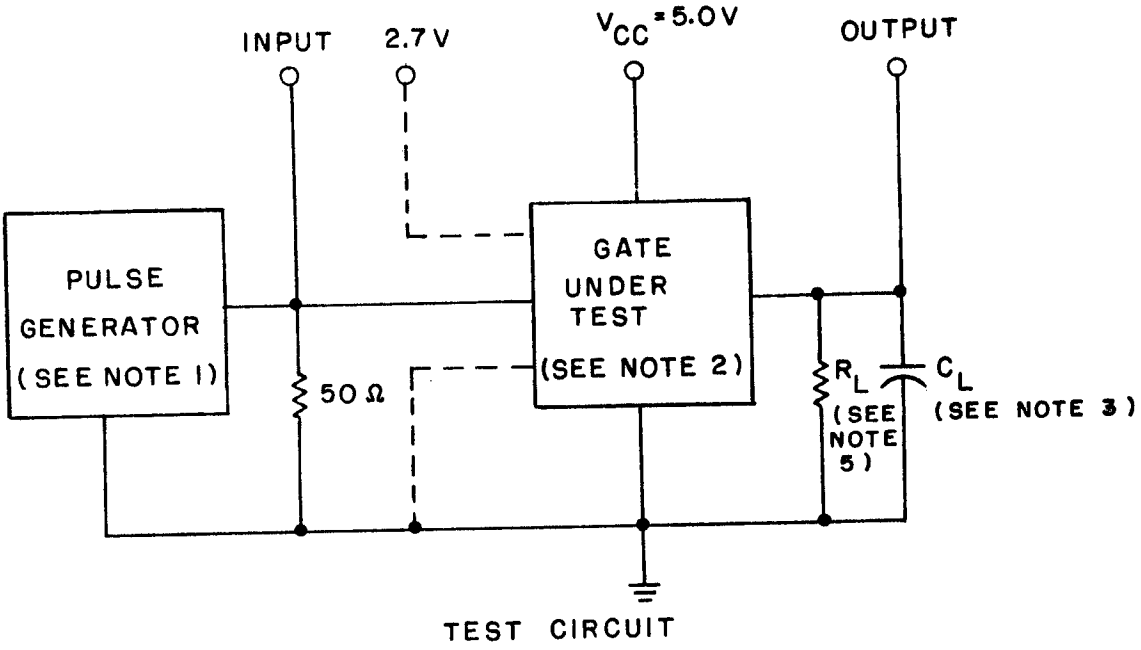
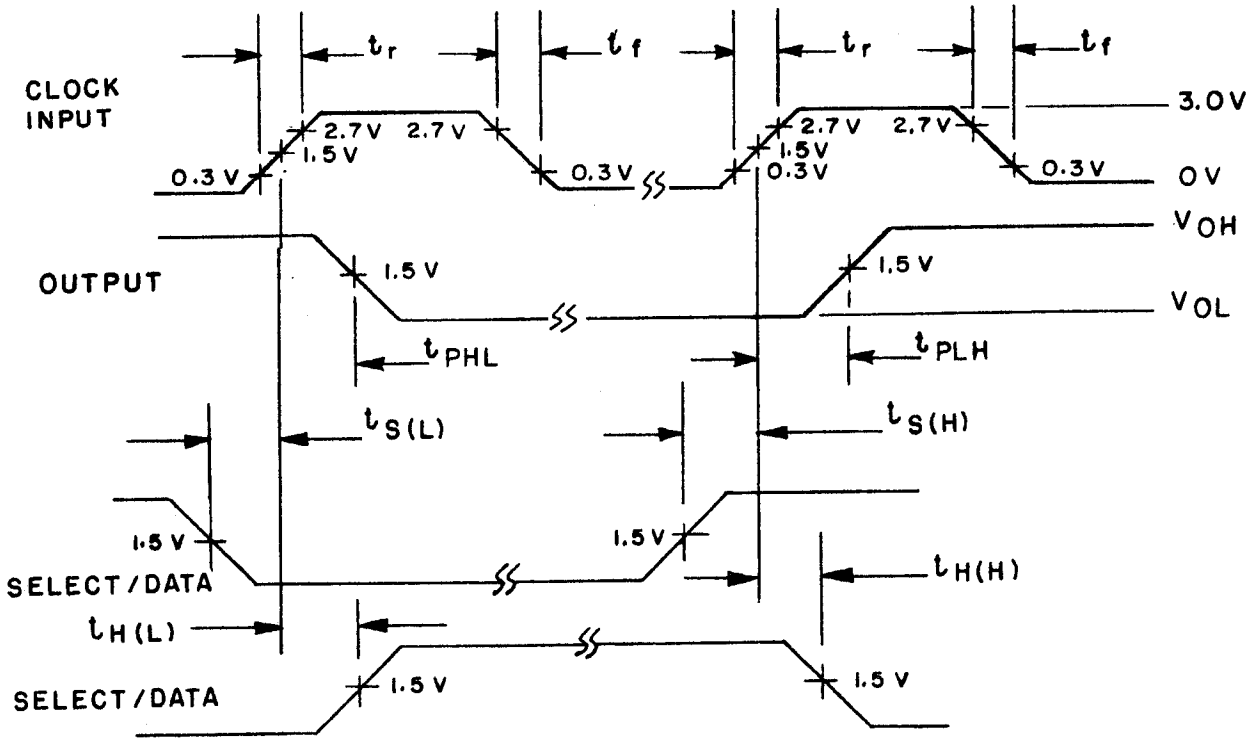


FIGURE 4. Switching test circuit.

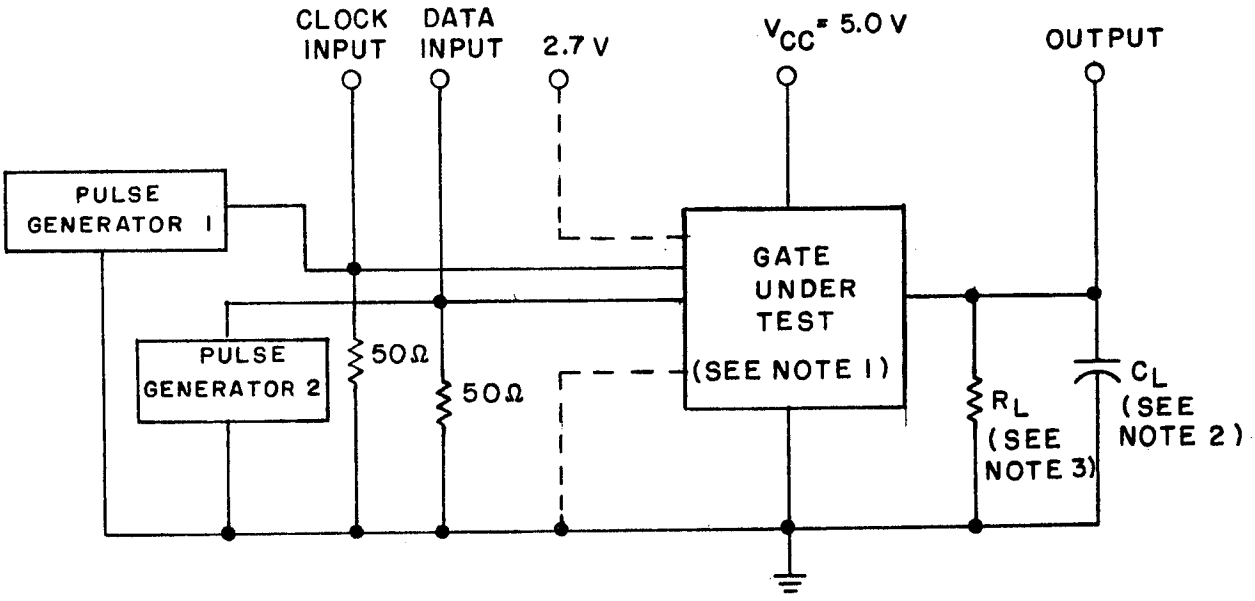
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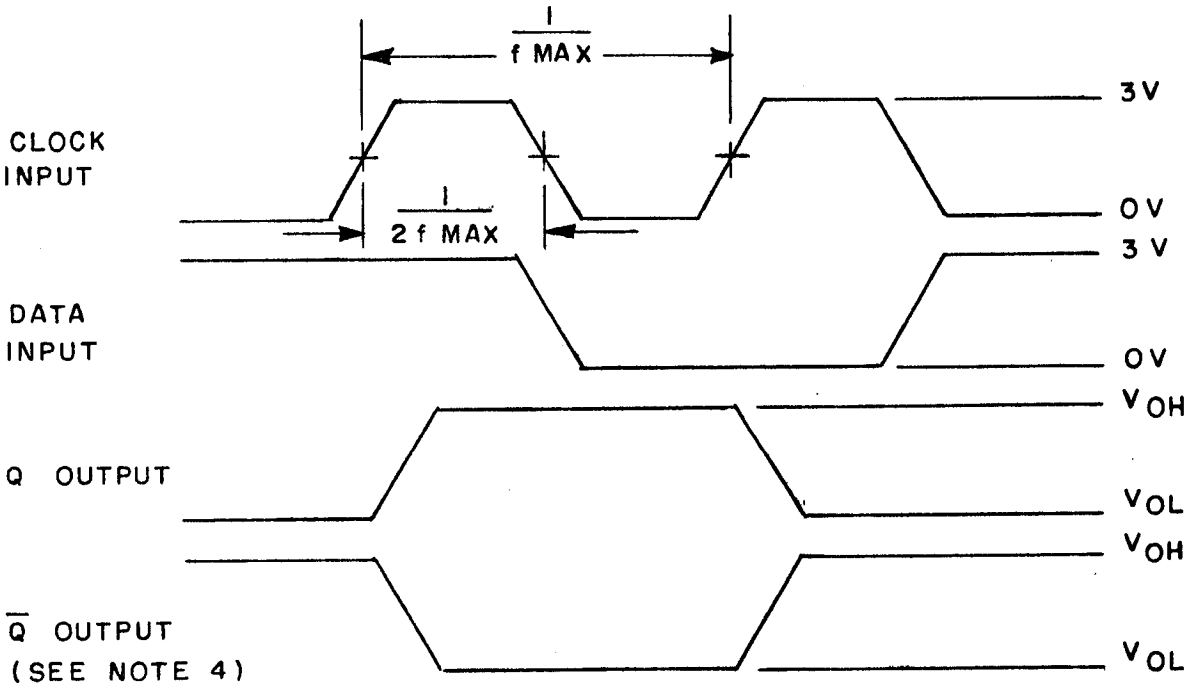
NOTES:
1. Pulse generator has following characteristics:
   $t_r = t_f \leq 2.5$ ns. PRR $\leq$ 1 MHz and $Z_{OUT} = 50\Omega$.
2. Inputs not under test are at 2.7 V or GND as specified in table III.
3. CL = 50 pF ±10% including scope, probe, wiring, and stray capacitance, without package in test fixture.
4. Voltage measurements are to be made with respect to network ground terminal.
5. $R_L = 500\Omega$ ±5%.
6. Setup and hold times, rising-edge clock for select input and data input.

FIGURE 4. Switching test circuit - Continued.

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TEST CIRCUIT



NOTES:
1. Inputs not under test are at 2.7 V or GND as specified in table III.
2. C_L = 50 pF ±10% including scope, probe, wiring, and stray capacitance, without package in test fixture.
3. R_L = 500Ω ±5%.
4. Device type 01 only.

FIGURE 5. Input clock frequency test circuit (all device types).

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be High ≥ 2.0 V, ≤ 0.8 V, or open)

Subgroups	Symbol	MIL-STD-883 method	Cases A, B, C, D, E, F, L, 2 test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Reserved terminal	Reserved terminal	Reserved terminal	Reserved terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{IH}	3006	1	-0.8 V	2.0 V	-1 mA							GND	$\overline{CP}$									V_{CC} = 4.5 V					2.5		V
			2				-1 mA																							
			3						-1 mA																					
			4								-1 mA																			
			5																											
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1 T_C = +25°C	V_{OL}	3007	33	-0.8 V	2.0 V	20 mA							GND	$\overline{CP}$									V_{CC} = 4.5 V						1.2	V
			34				20 mA																							
			35						20 mA																					
			36								20 mA																			
			37																											
			38																											
			39																											
			40																											
			41																											
			42																											
1 T_C = +25°C	V_{IC}		33	-0.8 V	2.0 V	-10 mA							GND	10 mA									V_{CC} = 4.5 V							V
			34				-10 mA																							
			35						-10 mA																					
			36								-10 mA																			
			37															-10 mA												
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			42																				-10 mA							

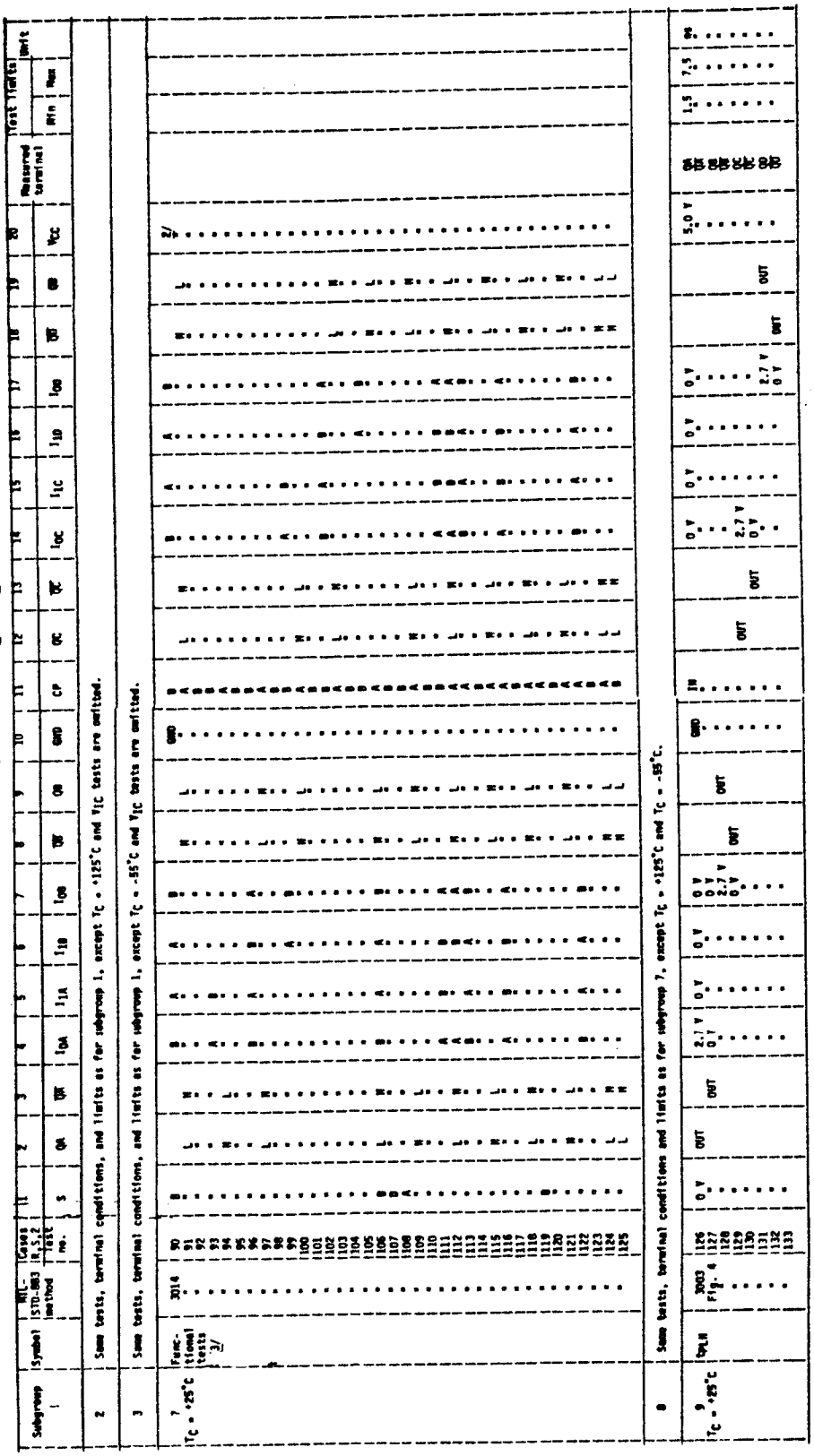
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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, C, D, E, F, L, 2 test no.	1 S	2 QA	3 QB	4 I_{QA}	5 I_{1A}	6 I_{1B}	7 I_{QB}	8 TBF	9 QB	10 GND	11 CP	12 QC	13 QC*	14 I_{QC}	15 I_{1C}	16 I_{1D}	17 I_{QD}	18 QD*	19 QB	20 V_{CC}	Reserved terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{IH1}	3010	43	2.7 V			2.7 V	2.7 V	2.7 V	2.7 V			GND	2.7 V			2.7 V	2.7 V	2.7 V	2.7 V			5.5 V	S I_{1A} I_{1B} I_{QB} I_{1C} I_{1D} I_{QD}		20	μA
			44																							
			45	0 V																						
			46	0 V																						
			47																							
			48											2.7 V			2.7 V	2.7 V	2.7 V	2.7 V						
			49																							
			50	0 V																						
			51	0 V																						
			52	0 V																						
	I_{IH2}		53	7.0 V			7.0 V	7.0 V	7.0 V	7.0 V				7.0 V			7.0 V	7.0 V	7.0 V	7.0 V				S I_{1A} I_{1B} I_{QB} I_{1C} I_{1D} I_{QD}			
			54																							
			55	0 V																						
			56	0 V																						
			57																							
			58											7.0 V			7.0 V	7.0 V	7.0 V	7.0 V						
			59																							
			60	0 V																						
			61	0 V																						
			62																							
	I_{IL}	3009	63	.5 V			.5 V	.5 V	.5 V	.5 V				.5 V			.5 V	.5 V	.5 V	.5 V				S I_{1A} I_{1B} I_{QB} I_{1C} I_{1D} I_{QD}			μA
			64	0 V																						
			65	5.5 V																						
			66	5.5 V																						
			67	0 V																						
			68	5.5 V																						
			69	0 V																						
			70	5.5 V																						
			71	5.5 V																						
			72	0 V																						
	I_{OS}	3011	73		0 V		5.5 V 0 V	5.5 V 0 V	0 V 5.5 V	0 V 5.5 V	0 V	0 V		1/			5.5 V 0 V	5.5 V 0 V	0 V 5.5 V	0 V 5.5 V	0 V				-60	-150	mA
			74																							
			75																							
			76																							
			77																							
			78																							
			79																							
			80																							
	I_{OD}		81		2.5 V		0 V 5.5 V	0 V 5.5 V	5.5 V 0 V	5.5 V 0 V	2.5 V	2.5 V			2.5 V	2.5 V	0 V 5.5 V	0 V 5.5 V	5.5 V 0 V	5.5 V 0 V	2.5 V	2.5 V	4.5 V		-60		mA
			82																							
			83		2.5 V																					
			84																							
			85																							
			86																							
			87																							
			88																							
	I_{CC}	3005	89				5.5 V	5.5 V	5.5 V	5.5 V							5.5 V	5.5 V	5.5 V	5.5 V			5.5 V	V_{CC}		30	mA

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, ≤0.8 V, or open)



Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				S	Q_A	$\overline{Q_B}$	I_{QA}	I_{1A}	I_{1B}	I_{QB}	$\overline{Q_B}$	Q_B	GND	CP	Q_C	$\overline{Q_C}$	I_{QC}	I_{1C}	I_{1D}	I_{QD}	$\overline{Q_D}$	Q_D	V_{CC}				
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																										
7 T_C = +25°C	Functional tests 3/	3014	90										GND										2/				
			91																								
			92																								
			93																								
			94																								
			95																								
			96																								
			97																								
			98																								
			99																								
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			118																								
			119																								
			120																								
			121																								
			122																								
			123																								
			124																								
			125																								
8	Same tests, terminal conditions and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																										
9 T_C = +25°C	t_{PLH}	3003 Fig. 4	126	0 V	OUT		2.7 V 0 V	0 V		0 V 0 V 2.7 V 0 V		OUT		IN			0 V	0 V	0 V	0 V			5.0 V		1.5	7.5	ns
			127																								
			128																								
			129																								
			130																								
			131																								
			132																								
			133																								

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases B,S,Z test no.	S	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = +25°C	I_{PLH}	3003 Fig. 4	134 135 136 137 138 139 140 141	2.7 V "	OUT	OUT	0.7 "	2.7 V 0 V "	0 V 0 V 2.7 V 0 V "	0 V 0 V 2.7 V 0 V "	OUT		GND	IR	OUT	OUT	0 V "	0 V "	0 V "	0 V "	OUT	OUT	5.0 V "		1.5 "	ns "
	I_{PHL}		142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157	0 V "	OUT	OUT	0.7 2.7 V "	0 V "	0 V "	2.7 V 0 V 2.7 V "	OUT		GND		OUT	OUT	0 V "	0 V "	2.7 V 0 V "	2.7 V 0 V "	OUT	OUT			9.5 "	
	I_{PSLZ}	3003 Fig. 5	158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173	0 V "	OUT	OUT	1MΩ 1MΩ 0 V "	0 V "	0 V "	0 V 0 V 1MΩ 1MΩ 0 V "	OUT		GND	IR1	OUT	OUT	0 V "	0 V "	0 V "	0 V "	OUT	OUT			100 "		MHz "
10	Same tests and terminal conditions as subgroup 9, except T_C = +125°C and use limits from table I.																										
11	Same tests and terminal conditions as subgroup 9, except T_C = -55°C and use limits from table I.																										

1/ Apply all voltages, then apply 3 V, 0 V, 3 V to CP then make measurement.
2/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.
3/ A = 3.0 V min, H greater than 1.5 V
B = 0.8 V or GND, L less than 1.5 V
4/ I_{IL} limits shall be as follows:

Test	MIL-STD-883 method	Limits (in mA)	
		Case X	Case U, Z
I_{IL}	-.25/--.80	-.02/--.00	

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TABLE III. Group A inspection for device types 02.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2 E, F Test no.	2 1 S	3 2 OA	4 3 I_{OA}	5 4 I_{1A}	7 5 I_{1B}	8 6 I_{OB}	9 7 OB	10 8 GND	12 9 CP	13 10 OC	14 11 I_{OC}	15 12 I_{1C}	17 13 I_{1D}	18 14 I_{OD}	19 15 OD	20 16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	.8 V	-1 mA	2.0 V	.8 V				GND	1/							4.5 V	S	2.5		V
	"	"	2	2.0 V	-1 mA	.8 V	2.0 V				"	"							"	I_{OA}	"		"
	"	"	3	.8 V				.8 V	2.0 V	-1 mA	"	"							"	I_{1A}	"		"
	"	"	4	2.0 V				2.0 V	.8 V	-1 mA	"	"							"	I_{1B}	"		"
	"	"	5	.8 V							"	"	-1 mA	2.0 V	.8 V				"	I_{OB}	"		"
	"	"	6	2.0 V							"	"	-1 mA	.8 V	2.0 V				"	OB	"		"
	"	"	7	.8 V							"	"				.8 V	2.0 V	-1 mA	"	I_{OC}	"		"
	"	"	8	2.0 V							"	"				2.0 V	.8 V	-1 mA	"	I_{1C}	"		"
																				I_{1D}			
																				I_{OD}			
																				OD			
	V_{OL}	3007	9	.8 V	20 mA	.8 V	2.0 V				"	"							"	S		.5	"
	"	"	10	2.0 V	20 mA	2.0 V	.8 V				"	"							"	I_{OA}		"	"
	"	"	11	.8 V				2.0 V	.8 V	20 mA	"	"							"	I_{1A}		"	"
	"	"	12	2.0 V				.8 V	2.0 V	20 mA	"	"							"	I_{1B}		"	"
	"	"	13	.8 V							"	"	20 mA	.8 V	2.0 V				"	I_{OB}		"	"
	"	"	14	2.0 V							"	"	20 mA	2.0 V	.8 V				"	OB		"	"
	"	"	15	.8 V							"	"				2.0 V	.8 V	20 mA	"	I_{OC}		"	"
	"	"	16	2.0 V							"	"				.8 V	2.0 V	20 mA	"	I_{1C}		"	"
																				I_{1D}			
																				I_{OD}			
																				OD			
	V_{IC}		17	-18 mA		-18 mA					"	"							"	S		-1.2	"
			18				-18 mA				"	"							"	I_{OA}		"	"
			19					-18 mA			"	"							"	I_{1A}		"	"
			20						-18 mA		"	"							"	I_{1B}		"	"
			21								"	"							"	I_{OB}		"	"
			22								"	-18 mA							"	OB		"	"
			23								"	"		-18 mA					"	I_{OC}		"	"
			24								"	"			-18 mA				"	I_{1C}		"	"
			25								"	"				-18 mA			"	I_{1D}		"	"
			26								"	"							"	I_{OD}		"	"
	I_{IH1}	3010	27	2.7 V		2.7 V					"	"							5.5 V	S		20	mA
	"	"	28				2.7 V				"	"							"	I_{OA}		"	"
	"	"	29	0 V				2.7 V			"	"							"	I_{1A}		"	"
	"	"	30	0 V					2.7 V		"	"							"	I_{1B}		"	"
	"	"	31								"	"							"	I_{OB}		"	"
	"	"	32								"	2.7 V							"	OB		"	"
	"	"	33								"	"		2.7 V					"	I_{OC}		"	"
	"	"	34	0 V							"	"			2.7 V				"	I_{1C}		"	"
	"	"	35	0 V							"	"				2.7 V			"	I_{1D}		"	"
	"	"	36								"	"					2.7 V		"	I_{OD}		"	"
																				OD			
	I_{IH2}	"	37	7.0 V		7.0 V					"	"							"	S		100	"
	"	"	38				7.0 V				"	"							"	I_{OA}		"	"
	"	"	39	0 V				7.0 V			"	"							"	I_{1A}		"	"
	"	"	40	0 V					7.0 V		"	"							"	I_{1B}		"	"
	"	"	41								"	"							"	I_{OB}		"	"
	"	"	42								"	7.0 V							"	OB		"	"
	"	"	43								"	"		7.0 V					"	I_{OC}		"	"
	"	"	44	0 V							"	"			7.0 V				"	I_{1C}		"	"
	"	"	45	0 V							"	"				7.0 V			"	I_{1D}		"	"
	"	"	46								"	"					7.0 V		"	I_{OD}		"	"
																				OD			
	I_{IL}	3009	47	.5 V		.5 V					"	"							"	S	4/	4/	mA
	"	"	48	0 V			.5 V				"	"							"	I_{OA}	"	"	"
	"	"	49	5.5 V				.5 V			"	"							"	I_{1A}	"	"	"
	"	"	50	5.5 V					.5 V		"	"							"	I_{1B}	"	"	"
	"	"	51	0 V							"	"							"	I_{OB}	"	"	"
	"	"	52	5.5 V							"	.5 V							"	OB	"	"	"
	"	"	53	0 V							"	"		.5 V					"	I_{OC}	"	"	"
	"	"	54	5.5 V							"	"			.5 V				"	I_{1C}	"	"	"
	"	"	55	5.5 V							"	"				.5 V			"	I_{1D}	"	"	"
	"	"	56	0 V							"	"					.5 V		"	I_{OD}	"	"	"
																				OD			

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TABLE III. Group A inspection for device types 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2, E, F Test no.	2 1 S	3 2 QA	4 3 I_{0A}	5 4 I_{1A}	7 5 I_{1B}	8 6 I_{0B}	9 7 QB	10 8 GND	12 9 CP	13 10 QC	14 11 I_{0C}	15 12 I_{1C}	17 13 I_{1D}	18 14 I_{0D}	19 15 QD	20 16 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{OS}	3011	57		0 V	5.5 V	5.5 V	5.5 V	5.5 V	0 V	GND	1/	0 V	5.5 V	5.5 V	5.5 V	5.5 V	0 V	5.5 V	9A	-60	-150	mA
			58																	9B			
			59																	9C			
			60																	9D			
	I_{OD}		61		2.5 V	0 V	0 V	0 V	0 V	2.5 V			2.5 V	0 V	0 V	0 V	0 V	2.5 V	4.5 V	9A			
			62																	9B			
			63																	9C			
			64																	9D			
	I_{CC}	3005	65			5.5 V	5.5 V	5.5 V	5.5 V					5.5 V	5.5 V	5.5 V	5.5 V		5.5 V	V_{CC}		34	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Functional tests 3/	3014	66	B	L	B	A	A	B	L	GND	B	L	B	A	A	B	L	2/				
			67									A											
			68									B											
			69		H	A	B	A	A	H		B	H	A	B	B	A	H					
			70									A											
			71									B											
			72		L	B	A	B	B	L		A	L	B	A	A	B	L					
			73									B											
			74									A											
			75					A		H		B	H	A	B	B	A	H					
			76									A											
			77									B											
			78									A	L	B	A	A	B	L					
			79									B											
			80									A											
			81									B	H	A	B	B	A	H					
			82									A											
			83									B											
			84	A								A	L	B	A	A	B	L					
			85		H					H		B	H	A	B	B	A	H					
			86									A											
			87			A	B	B	A	L		B	H	A	B	B	A	H					
			88									A											
			89		L	B	A	B	B	H		B	L	A	B	B	A	L					
			90									A											
			91		H	A	B	A	A	L		B	H	A	B	B	A	H					
			92									A											
			93									B	L	B	A	A	B	L					
			94									A											
			95	B	L					H		B	H	A	B	B	A	H					
			96									A											
			97		H	B	A	A	B	L		B	L	A	B	B	A	L					
			98									A											
			99		L	A	B	B	A	H		B	H	A	B	B	A	H					
			100									A											
			101		L	B	A	A	B	L		B	L	A	B	B	A	L					
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																						

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TABLE III. Group A inspection for device types 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 2 E, F Test no.	2 1	3 2	4 3	5 4	7 5	8 6	9 7	10 8	12 9	13 10	14 11	15 12	17 13	18 14	19 15	20 16	Measured terminal	Test limits Min	Test limits Max	Unit
				S	QA	I_{OA}	I_{1A}	I_{1B}	I_{OB}	QB	GND	CP	QC	I_{OC}	I_{1C}	I_{1D}	I_{OD}	QD	V_{CC}				
9 T_C = +25°C	t_{PLH}	3003 Fig. 4	102	0 V	OUT	2.7 V	0 V	0 V	0 V	OUT	GND	IN		0 V	0 V	0 V	0 V		5.0 V	QA	1.5	7.5	ns
			103	"		"	"	"	2.7 V		"	"		0 V	"	"	"		"	QB	"	"	"
			104	"		"	"	"	"		"	"	OUT	2.7 V	"	"	"		"	QC	"	"	"
			105	"		0 V	"	"	"		"	"		2.7 V	"	"	2.7 V	OUT	"	QD	"	"	"
			106	2.7 V	OUT	"	2.7 V	"	0 V		"	"		0 V	"	"	0 V		"	QA	"	"	"
			107	"		"	"	2.7 V	"	OUT	"	"		"	2.7 V	"	"		"	QB	"	"	"
			108	"		"	"	"	"		"	"	OUT	"	2.7 V	2.7 V	"		"	QC	"	"	"
			109	"		"	"	"	"		"	"		"	"	"	"	OUT	"	QD	"	"	"
	t_{PHL}		110	0 V	OUT	0 V	0 V	0 V	2.7 V	OUT	"	IN		2.7 V	0 V	0 V	2.7 V		"	QA	"	9.5	"
			111	"		"	"	"	0 V		"	"		2.7 V	"	"	"		"	QB	"	"	"
			112	"		"	"	"	"		"	"	OUT	0 V	"	"	"		"	QC	"	"	"
			113	0 V		2.7 V	"	"	"		"	"		"	2.7 V	2.7 V	0 V	OUT	"	QD	"	"	"
			114	2.7 V	OUT	0 V	"	2.7 V	"		"	"		"	2.7 V	"	"		"	QA	"	"	"
			115	2.7 V		0 V	"	0 V	"	OUT	"	"		"	0 V	"	"		"	QB	"	"	"
			116	2.7 V		0 V	"	"	"		"	"	OUT	"	0 V	0 V	"	OUT	"	QC	"	"	"
			117	2.7 V		0 V	"	"	"		"	"		"	"	"	"		"	QD	"	"	"
	f_{MAX}	3003 Fig. 5	118	0 V	OUT	IN2	0 V	0 V	0 V	OUT	"	IN1		0 V	0 V	0 V	0 V		"	QA	100		MHz
			119	"		0 V	"	"	IN2		"	"		0 V	"	"	"		"	QB	"		"
			120	"		"	"	"	0 V		"	"	OUT	IN2	"	"	IN2	OUT	"	QC	"		"
			121	"		"	"	"	"		"	"		0 V	"	"	0 V		"	QD	"		"
			122	2.7 V	OUT	"	IN2	0 V	"		"	"		"	"	"	"		"	QA	"		"
			123	"		"	0 V	IN2	"	OUT	"	"		"	IN2	"	"		"	QB	"		"
			124	"		"	"	0 V	"		"	"	OUT	"	0 V	IN2	"	OUT	"	QC	"		"
			125	"		"	"	0 V	"		"	"		"	"	"	"		"	QD	"		"
10	Same tests and terminal conditions as subgroup 9, except T_C = +125°C and use limits from table I.																						
11	Same tests and terminal conditions as subgroup 9, except T_C = -55°C and use limits from table I.																						

NOTES:

1/ Apply all voltages, then apply 3 V, 0 V, 3 V to CP then make measurements.

2/ Perform functional sequence at V_{CC} = 4.5 V and 5.5 V both.

3/ A = 3.0 V min, B = 0 V or GND, H > 1.5 V, and L < 1.5 V.

4/ I_{IL} limits shall be as follows:

Test	Min/Max limits in mA		
	CKT A	B	C
I_{IL}	-.25/-.60	-.03/-.60	

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4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
I_{IN}	Current flowing into an input terminal.
V_{IN}	Voltage level at an input terminal

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F398
02	54F399

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6.6 Manufacturers' designations. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuits		
	A	B	C
	Fairchild	Motorola Inc.	Signetics Corp.
01			
02			

Custodians:
Air Force - 17
Navy - EC
Army - ER

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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